The Proton So Far: an emergent color singlet, its observables, and lattice convergence

Tessera — cobordism subsystem (#398 / #400 / #404, under the #380 epic)

Abstract

This is a status report on building a proton from quarks in the tessera cobordism/Regge subsystem. Three results are in hand. (1) **The emergent color singlet** (#398): on the trivalent junction W_{ABC} with *symmetrically* placed register windows, the input→result transport intertwines the color $\mathbb{Z}_3$, and a natural color-symmetric quark input transports to the net-color-zero singlet line $\langle(1, \omega, \omega^2)\rangle$ with overlap 0.999 — the output's location is forced by symmetry, not hand-placed. (2) **Emergent observables** (#400), read off the relaxed singlet: the color charge $\sigma \to 0$ (confinement), a length scale, and a (proxy) mass, giving the dimensionless $r \cdot m \approx 6.9$ (the physical proton's value is ≈ 4.0). (3) **Lattice convergence** (#404): the granularity is tunable (a frequency-N geodesic base); refining it shrinks the intertwiner residual while the singlet overlap sharpens to 1 — consistent with the small imperfections being discretization artifacts, not symmetry breaking. The honest assessment: we have the *color-confinement skeleton* of a baryon — emergent, structurally correct, with a scale-free observable in the right ballpark — on a coarse lattice; not yet a quantitatively faithful proton (the size is still lattice-pinned, the mass is a curvature proxy, and the electroweak quantum numbers are absent). All numbers are measured through the production C++ `TransportCobordism`.

Contents

1 The program

The proton-from-quarks program relaxes a Lorentzian Regge cobordism with pinned boundary color states and reads the *emergent* result off the relaxed geometry — nothing is imposed. The color singlet of $SU(3)$ is the alternating invariant ε_{ijk}, which is absent from $\mathbf{3} \otimes \mathbf{3}$ and first appears in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3}$: a baryon is irreducibly a three-quark object, so pairwise (bipartite) merges cannot reach it (#382). The resolution is the trivalent junction W_{ABC} (#396): one connected geodesic icosahedron (S^2) minus twelve vertex-disjoint hole triangles in four *windows* of three — the quark inputs A, B, C and the emergent result R — extruded $\times I$. Distinct holes are independent cycles ($b_1 = 11$), so the inputs do not average, and charge is conserved at the junction by the global Stokes relation $\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$.

2 The emergent color singlet (#398)

The four windows are placed *symmetrically*: A_4, a tetrahedral subgroup of the icosahedral rotation group, partitions one tetrahedral set of icosahedron vertices into four blocks of three (the window corners), and permutes the four blocks transitively — so the windows are A_4-equivalent. Write g for the A_4 three-cycle that fixes R and cycles $A \to B \to C$; on the result holes it is the cyclic color $\mathbb{Z}_3$ (the cyclic subgroup of the hole-permutation S_3/Weyl symmetry, which is why only the cube roots of unity appear). Let P_{in} (on the 9-dimensional input space over A, B, C) and P_{out} (on the 3-dimensional result space over R's holes) be its signed permutation action on hole periods, sharing an orientation convention, and let $M : V_{\text{in}} \to V_{\text{out}}$ be the input→result transport. Then M **intertwines** the symmetry, to the residual of Table 2,

$$M\, P_{\text{in}} \approx P_{\text{out}}\, M. \tag{1}$$

P_{out} is the color $\mathbb{Z}_3$ (eigenvalues $\{1, \omega, \omega^2\}$); its ω-eigenline is the singlet direction $\langle(1, \omega, \omega^2)\rangle$. This is the discrete *net-color-zero* ($\sigma = 0$) surrogate for the SU(3) singlet — it is the confinement condition (orthogonality to the charge mode $(1, 1, 1)$), the $\mathbb{Z}_3$ shadow of ε_{ijk}, not literally ε_{ijk} (which is slot-permutation invariant). By (1), the ω-eigenline of P_{in} — the natural color-symmetric quark input — is carried into the singlet line up to the residual: the output's location is *forced* (not hand-placed), even though the input representation is the symmetry-distinguished choice.

Three ingredients are jointly necessary (symmetric windows alone are not enough): a symmetry-respecting metric, the input in the ω-representation of g, and a g-coherent signed read-out. With them, the measured overlap is 0.999 (the 0.001 deficit is the intertwiner residual; a greedy, geometrically inequivalent placement reaches only ~ 0.74). Details are in the companion note *The Proton Color Singlet from Symmetric Register Windows.*

3 Emergent observables (#400)

With the singlet in hand we read the proton's observables *off the relaxed geometry*, emergent-first, reusing the existing measurement methods. The singlet survives relaxation (0.995), so all reads come from one relaxed geometry.

observable	value	how it is read
color charge σ	$0.06 \approx 0$	projection onto P_{out}'s $+1$ (charge) mode — confinement
singlet component	0.995	the ω-mode (the proton)
radius r (RMS edge / lattice scale)	1.063	$\sqrt{\langle \ell^2 \rangle_{\ell^2 > 0}}$ over relaxed spacelike edges
mass B (curvature anchor)	6.53	shell-summed Re-deficit (the #352 method)
mass A (alt. proxy)	228.8	|dual Regge action|
$r{\cdot}m$ (anchor B)	**6.94**	vs the physical proton's ≈ 4.0
$r{\cdot}m$ (alt. proxy A)	243	

The color charge is the g-invariant ($+1$) mode of the result; the singlet is the orthogonal ω-mode, so $\sigma \to 0$ is confinement — this realizes the Stokes relation $\sigma_R = -(\sigma_A + \sigma_B + \sigma_C)$, a color-symmetric input summing to zero result charge. **The two masses are different, un-normalized**

proxies — a dual-action magnitude (A) versus a shell-summed real deficit (B) — and are not expected to agree; the $\sim 35\times$ gap is a normalization difference, not a consistency check. B is the established #352 curvature anchor used below; A is reported for scale only. The dimensionless $r \cdot m$ is the scale-free observable to compare across unit systems: the physical proton has $r_p m_p \approx (0.84 \text{ fm})(938 \text{ MeV}/\hbar c) \approx 4.0$ — its radius is about four reduced Compton wavelengths. Our 6.9 (anchored on the curvature mass B) is the same order of magnitude, and a large improvement on the prior crude ~ 73 from the bare-quark merge (#352); but the comparison is heuristic (proxies; see §5).

4 Lattice convergence (#404)

The base granularity is tunable: a frequency-N geodesic icosahedron (`set_frequency(N)`, $N = 2$ the 42-vertex default), with the same A_4-orbit windows generated from the symmetry at any N. The small imperfection in (1) is a fixed-resolution triangulation artifact (the $\times I$ extrusion is not g-symmetric even though the base surface and metric are), so it should shrink with resolution while the singlet overlap sharpens toward 1:

N	cobordism vertices	$\lVert MP_{\text{in}} - P_{\text{out}}M \rVert \,/\, \lVert M \rVert$	singlet overlap
2	126	4.26×10^{-2}	0.9991–0.9998
3	276	2.09×10^{-2}	0.9996–0.99999
4	486	1.69×10^{-2}	0.9997–1.0000

The overlap is essentially exact at every N; the residual decreases monotonically, but the $N{=}3 \to 4$ step is small ($\sim$19%), so whether it reaches zero or floors near $\sim 1.7{\times}10^{-2}$ is not yet established. The strong evidence that the imperfection is a discretization artifact, not symmetry breaking, is the overlap $\to 1$. $N = 2$ reproduces the #398 base exactly (backward-compatible: the junction suite passes 20/20, and `set_frequency` rejects $N < 2$, the 42-vertex floor).

5 Are we looking at a proton?

A careful answer: we have the **color-confinement skeleton** of a baryon, not a quantitatively faithful proton.

What is genuinely real. The color singlet. Three colored quarks bind into a color-neutral, confined state — the defining property of a baryon — and its *location* emerges from a symmetric input, forced by the geometry's $\mathbb{Z}_3$ symmetry, never hand-placed ($\sigma \to 0$, singlet 0.995, robust through relaxation and sharpening with resolution).

What is still a proxy. The *radius* is essentially the lattice scale (the RMS edge length $r \approx 1.06$ on the uniform seed): the geometry did not develop its own localized size, so r carries little physical information yet, and $r \cdot m$ is dominated by the mass term. The *mass* is a curvature (deficit-angle) proxy, not a calibrated rest mass. So the 6.9 vs 4.0 agreement is an order-of-magnitude correspondence for a proxy, not a measurement. And this is only the *color* sector: the proton is also electric charge +1, spin-$\frac{1}{2}$, and flavor *uud*, none of which this yet captures; the lattice is also coarse.

Verdict. A proton-shaped object: the right confinement and color-singlet structure, arrived at emergently with the correct symmetry, and a scale-free observable in the right ballpark — the QCD skeleton of a proton on a coarse lattice, not the full particle.

6 Outlook

Three follow-ons (the #380 epic) would push it toward "actual":

- **#403 — let the size emerge.** Relax from a localizing, g-invariant seed (the van Raamsdonk / entanglement seed, available via `set_entangled_metric`) so a genuine bound-state length scale emerges and the radius is no longer lattice-pinned; then $r \cdot m$ becomes a clean probe. This composes with the finer lattice (#404). (A reconnaissance confirms the entanglement seed develops a strong interior/exterior contrast where the uniform seed stays flat.)
- **#405 — electroweak quantum numbers.** Read the emergent electric charge +1 (the uud charges $+\frac{2}{3}, +\frac{2}{3}, -\frac{1}{3}$), then spin and flavor.
- **#361 — animation.** Visualize the relaxation: primal/dual geometry, curvature gradient, null (photon) edges, live residuals.